

# Track 1 — Chapter Build Pass

MPEP §§2103–2106 · Patent Eligibility Under 35 U.S.C. §101

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This is a PILOT section — built from your uploaded MPEP to test the format before the rest of Chapter 2100 and the other high-yield chapters are built out. Format: plain-English rule card → one worked example → one quiz question with answer, per rule. A chapter close-out practice set and a memorize-this list follow at the end. Read entirely offline — no interaction required.

Accuracy note: content is drawn directly from your uploaded MPEP file (current as of its revision date), translated into plain English. Spot-check anything with a specific legal test or citation against your MPEP if precision matters for a specific question.

## Overview: How Eligibility Fits Into Examination

Before an examiner looks at novelty (§102), obviousness (§103), or the written-description/enablement requirements (§112), they check whether the claim is even the kind of thing a patent can cover at all. That threshold check is §101 eligibility. It's a gate, not a merit review — a claim can be 100% novel and non-obvious and still get rejected for being ineligible subject matter (e.g., an abstract idea or a law of nature). This is a favorite exam distinction: eligibility (§101) is a completely separate inquiry from novelty (§102) and obviousness (§103), even though all three can appear in the same office action.

### Rule 1: The Four Statutory Categories (§2104)

#### PLAIN ENGLISH

To be eligible at all, a claim must fall into one of exactly four categories Congress listed in §101: a **process** (a method/series of steps), a **machine** (a device with moving/interacting parts), a **manufacture** (an article made by hand or machine that isn't a machine itself), or a **composition of matter** (a chemical compound, mixture, or combination of substances). If a claim doesn't fit any of these four buckets, it fails eligibility immediately — no need to even reach the abstract-idea/law-of-nature analysis.

#### EXAMPLE

A claim to "a computer-readable medium storing instructions that cause a processor to sort a list" is a manufacture (the physical medium) — eligible category-wise. But a claim to just "a signal encoding sorted data" (with no physical medium) has been found NOT to fit any of the four categories, since a transitory signal isn't a process, machine, manufacture, or composition of matter — an automatic §101 rejection before the abstract-idea analysis is even needed.

#### QUIZ

A claim is drawn to "a transitory propagating signal carrying encoded data." What's the correct basis for rejecting it under §101?

- (A) It's an abstract idea under Step 2A
- (B) It fails to recite one of the four statutory categories
- (C) It lacks a practical application under Step 2A Prong Two
- (D) It's not eligible for lack of novelty

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### Rule 2: The Overall Eligibility Framework: Step 1 → Step 2A → Step 2B (§2106)

#### PLAIN ENGLISH

Examiners follow a fixed sequence (the Alice/Mayo framework, as implemented in the USPTO's eligibility guidance): **Step 1** — does the claim fall into one of the four statutory categories (Rule 1 above)? If no, reject immediately. If yes, go to **Step 2A**, which has two prongs. **Prong One**: does the claim recite a judicial exception (an abstract idea, law of nature, or natural phenomenon)? If no, the claim is eligible — stop here. If yes, go to **Prong Two**: is that exception integrated into a "practical application"? If yes, the claim is eligible — stop here. If no, proceed to **Step 2B**: does the claim add "significantly more" than the exception itself (an inventive concept)? If yes, eligible. If no, reject under §101.

### EXAMPLE

A claim recites "a method of calculating a discount using the formula  $X = Y \times 0.9$ ." Step 1: it's a process — pass. Step 2A Prong One: a mathematical formula is an abstract idea — exception found. Prong Two: nothing in the claim ties the calculation to any practical use (no display, no application to a physical process) — no practical application. Step 2B: nothing else in the claim adds an inventive concept — it's just the bare formula. Result: rejected under §101.

### QUIZ

**A claim passes Step 1 and Step 2A Prong One finds an abstract idea. At Prong Two, the examiner determines the abstract idea IS integrated into a practical application. What happens next?**

- (A) Proceed to Step 2B to look for an inventive concept
- (B) The claim is eligible; the analysis stops
- (C) The claim is automatically rejected
- (D) Restart at Step 1 with the practical application in mind

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## Rule 3: Step 2A Prong One — The Three Abstract Idea Groupings (§2106.04(a))

### PLAIN ENGLISH

Not every abstract concept counts. USPTO guidance narrows "abstract ideas" to three groupings: **(1) Mathematical concepts** — formulas, equations, calculations, relationships. **(2) Certain methods of organizing human activity** — fundamental economic principles, commercial or legal interactions (contracts, hedging, insurance), managing personal behavior or relationships. **(3) Mental processes** — concepts performed in the human mind (observation, evaluation, judgment, opinion) — including ones merely recited as being done "with a computer" if a human could still do it mentally. If a claim limitation doesn't fall into one of these three groupings, examiners generally should NOT treat it as an abstract idea, even if it feels abstract intuitively.

### EXAMPLE

A claim to "determining a risk level by comparing a patient's data to a threshold and displaying a color-coded alert" — the "comparing... to a threshold" step is a mental process (an observation/evaluation a human could do in their head), even though the claim recites a computer. The "displaying a color-coded alert" step is not itself abstract, but that doesn't matter yet — Prong One just asks if ANY limitation recites an abstract idea, and the comparison step does.

### QUIZ

**A claim recites steps for "negotiating a sales contract's terms based on market conditions." Which grouping does this most likely fall under?**

- (A) Mathematical concepts
- (B) Certain methods of organizing human activity
- (C) Mental processes
- (D) None — this isn't an abstract idea

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## Rule 4: Step 2A Prong One — Laws of Nature, Natural Phenomena & Products of Nature (§2106.04(b))

## PLAIN ENGLISH

The other two judicial exceptions: a **law of nature** (e.g.,  $E=mc^2$ , the relationship between a drug dose and its metabolite level in blood) and a **natural phenomenon**, which includes naturally occurring products. For a product claim to a natural material (a gene, mineral, organism), the test is whether it has "**markedly different characteristics**" from what exists in nature — a different structure, function, or property that isn't just a trivial or expected difference. If it does, it's not treated as a natural phenomenon exception at all.

## EXAMPLE

A claim to "an isolated DNA sequence identical to a naturally occurring gene" — merely isolating it doesn't create markedly different characteristics from the natural gene, so it's treated as a natural phenomenon exception (this mirrors the real case, *Assoc. for Molecular Pathology v. Myriad*). But a claim to a "synthetic cDNA" (missing the introns present in the natural gene) has a different chemical structure — markedly different — so it's not subject to the natural-phenomenon exception at all.

## QUIZ

**A claim covers a bacterium engineered to include a plasmid not found in any naturally occurring bacterium, giving it a new capability. Under the markedly different characteristics analysis, this claim is:**

- (A) A natural phenomenon, because bacteria occur in nature
- (B) Not subject to the natural-phenomenon exception, because the new capability is a markedly different characteristic
- (C) An abstract idea
- (D) Ineligible regardless of markedly different characteristics

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## Rule 5: Step 2A Prong Two — Integration Into a Practical Application (§2106.04(d))

### PLAIN ENGLISH

Once a judicial exception is found, Prong Two asks whether other claim elements integrate it into a practical application — i.e., apply, rely on, or use the exception in a way that imposes a meaningful limit, rather than just append it to the exception ("apply it" language, generic computer implementation, or insignificant extra-solution activity like mere data gathering don't count). Things that DO tend to show integration: improving the functioning of a computer or another technology, applying the exception to effect a particular treatment for a disease, or using a particular machine that's integral to the claim.

### EXAMPLE

A claim to a mental process ("detecting an irregular heartbeat pattern by comparing it to known patterns") that ALSO recites "and in response, delivering an electric pulse via an implanted pacemaker to correct the irregularity" — the pacemaker treatment step integrates the abstract mental comparison into a practical application (a specific treatment), making the claim eligible at Prong Two.

## QUIZ

A claim recites an abstract idea followed only by "apply the above using a generic computer processor." Is this integrated into a practical application?

- (A) Yes — using a computer is always sufficient
- (B) No — generic computer implementation alone doesn't integrate an exception
- (C) Yes, but only if the computer is described as "specialized"
- (D) This analysis doesn't apply to computer-implemented claims

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## Rule 6: Step 2B — Significantly More / Inventive Concept (§2106.05)

### PLAIN ENGLISH

If Prong Two doesn't find a practical application, examiners move to Step 2B: does the claim, considered as an ordered combination, add something beyond the exception itself that amounts to an inventive concept? Simply reciting **well-understood, routine, conventional activity** already known in the industry does NOT satisfy Step 2B — even if that activity wasn't disclosed in the prior art the examiner happens to cite. This is a lower bar than novelty, but it's not nothing: it looks for something beyond generic, conventional implementation.

### EXAMPLE

A claim to an abstract data-encryption algorithm that adds only "receiving data over a network" and "storing the result in memory" — both are well-understood, routine, conventional computer functions (the guidance gives these almost as textbook examples), so Step 2B is not satisfied and the claim remains ineligible.

## QUIZ

At Step 2B, an examiner rejects a claim because its only additional elements are conventional data-storage steps. The applicant argues no prior art reference discloses that specific storage method. Is this argument persuasive against the §101 rejection?

- (A) Yes — if it's not in the prior art, it can't be "well-understood, routine, conventional"
- (B) No — well-understood/routine/conventional is evaluated independently of whether prior art discloses it
- (C) Yes, but only for software claims
- (D) No — Step 2B doesn't consider claim elements at all

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## Rule 7: Streamlined Eligibility Analysis (§2106.06)

### PLAIN ENGLISH

For claims that clearly do not raise any substantial eligibility concern on their face, examiners may use a streamlined analysis and skip the full Step 2A/2B write-up — e.g., a claim to a complex, specific manufactured device (a new kind of engine) plainly isn't an attempt to tie up an abstract idea or natural law, so a full eligibility rejection analysis isn't required. This doesn't change the legal standard — it's a shortcut for obvious cases, not a separate, looser test.

### EXAMPLE

A claim to "a mechanical gearbox comprising a specific arrangement of interlocking gears with defined tooth ratios" would typically qualify for streamlined analysis — no reasonable read of the claim suggests it's directed to an abstract idea or law of nature, so the examiner doesn't need to walk through the full Alice/Mayo write-up.

### QUIZ

**Streamlined eligibility analysis means:**

- (A) A different, more lenient eligibility standard applies
  - (B) A shortcut in the examiner's write-up for claims with no plausible eligibility issue, using the same underlying standard
  - (C) Automatic allowance without any §101 consideration
  - (D) A process only available for software claims
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## Chapter Close-Out: Mixed Practice Set

Mixing all seven rules above, in exam-style fact patterns. Answer key follows at the end of this set.

**1. A claim recites "a diagnostic method comprising: (a) measuring a biomarker level in a blood sample, and (b) determining that a level above 5 ng/mL indicates disease X." Nothing further is claimed. Under Step 2A Prong One, this claim most likely recites:**

- (A) A mathematical concept
- (B) A law of nature
- (C) A method of organizing human activity
- (D) No judicial exception

**2. Same claim as above. At Step 2A Prong Two, is the law of nature integrated into a practical application?**

- (A) Yes, because measuring blood is a physical step
- (B) No — merely detecting a natural correlation without acting on it isn't integration
- (C) Yes, because a specific numeric threshold is used
- (D) Cannot be determined without Step 2B

**3. A claim to "a new alloy composition consisting of 60% titanium, 30% nickel, and 10% cobalt, with a specific crystalline structure not found in nature" would most likely:**

- (A) Fail Step 1 for not being a statutory category
- (B) Qualify for streamlined eligibility analysis
- (C) Be rejected as an abstract idea
- (D) Require a full Step 2B inventive-concept analysis regardless

**4. True or False: A claim can be found eligible under §101 and still be rejected under §102 for lack of novelty.**

- (A) True
- (B) False

**5. A claim recites "a method of hedging risk in commodities trading by calculating a fixed bill amount using historical pricing data." This claim combines which two abstract idea groupings?**

- (A) Mental processes and laws of nature
- (B) Mathematical concepts and certain methods of organizing human activity
- (C) Products of nature and mental processes
- (D) None — hedging is not abstract

**6. At Step 2B, an examiner concludes the only additional claim element beyond the abstract idea is "transmitting the result to a remote server for storage." This is best characterized as:**

- (A) A meaningful inventive concept
- (B) Insignificant extra-solution activity / well-understood, routine, conventional activity
- (C) A new statutory category
- (D) A markedly different characteristic

**7. A claim fails Step 1 because it's directed to a transitory signal. What is the correct next step?**

- (A) Proceed to Step 2A Prong One
- (B) Reject under §101 immediately — no need to reach Step 2A/2B
- (C) Proceed directly to Step 2B
- (D) Analyze under §102 instead



## Memorize This: Key Tests & Terms

- **Four statutory categories:** process, machine, manufacture, composition of matter — a claim must fit ONE to even start the analysis.
- **The sequence:** Step 1 (category) → Step 2A Prong One (judicial exception?) → Step 2A Prong Two (practical application?) → Step 2B (significantly more?). Stop the moment you get an eligible result.
- **Three abstract idea groupings:** mathematical concepts; certain methods of organizing human activity; mental processes.
- **Markedly different characteristics test** — used for natural products/materials to see if the exception even applies.
- **Practical application** ≠ "apply it" or **generic computer implementation** — that language does NOT integrate an exception.
- **Well-understood, routine, conventional activity** fails Step 2B — independent of whether it's disclosed in prior art (§101 ≠ §102).
- **Streamlined analysis** = same legal standard, shortcut write-up, for claims with no plausible eligibility issue.
- **§101 eligibility is a threshold, standalone inquiry** — separate from §102 novelty, §103 obviousness, and §112 written description/enablement.

## Answer Key

All quiz answers for this PDF, in one place. Rule-card quizzes first, then the chapter close-out set.

### Rule-Card Quizzes

#### Rule 1: The Four Statutory Categories (§2104)

A claim is drawn to "a transitory propagating signal carrying encoded data." What's the correct basis for rejecting it under §101?

Answer: B

Transitory signals don't fit process, machine, manufacture, or composition of matter — this is a category failure, resolved before you'd ever reach the judicial-exception (abstract idea) analysis. Choice (D) is a trap: novelty is a §102 issue, unrelated to eligibility.

#### Rule 2: The Overall Eligibility Framework: Step 1 → Step 2A → Step 2B (§2106)

A claim passes Step 1 and Step 2A Prong One finds an abstract idea. At Prong Two, the examiner determines the abstract idea IS integrated into a practical application. What happens next?

Answer: B

If Prong Two finds integration into a practical application, the claim is eligible and you stop — Step 2B is only reached when Prong Two does NOT find a practical application. This order-of-operations point (stop as soon as you get an eligible result) is a common trap.

#### Rule 3: Step 2A Prong One — The Three Abstract Idea Groupings (§2106.04(a))

A claim recites steps for "negotiating a sales contract's terms based on market conditions." Which grouping does this most likely fall under?

Answer: B

Negotiating contract terms is a commercial/legal interaction — squarely within "certain methods of organizing human activity." Watch for these three labels; exam questions often just change the fact pattern's flavor (insurance, hedging, personal relationship management) while testing the same grouping.

#### Rule 4: Step 2A Prong One — Laws of Nature, Natural Phenomena & Products of Nature (§2106.04(b))

A claim covers a bacterium engineered to include a plasmid not found in any naturally occurring bacterium, giving it a new capability. Under the markedly different characteristics analysis, this claim is:

Answer: B

This mirrors *Diamond v. Chakrabarty* — a bacterium engineered with new, non-naturally-occurring capabilities has markedly different characteristics from anything in nature, so the natural-phenomenon exception doesn't even apply; the claim moves forward in the analysis without being tagged as an exception.

### **Rule 5: Step 2A Prong Two — Integration Into a Practical Application (§2106.04(d))**

A claim recites an abstract idea followed only by "apply the above using a generic computer processor." Is this integrated into a practical application?

Answer: B

Merely instructing a generic computer to implement an abstract idea is the classic Alice Corp. pattern the guidance explicitly rejects as integration — it's treated the same as "apply it." This is one of the most frequently tested traps in this section.

### **Rule 6: Step 2B — Significantly More / Inventive Concept (§2106.05)**

At Step 2B, an examiner rejects a claim because its only additional elements are conventional data-storage steps. The applicant argues no prior art reference discloses that specific storage method. Is this argument persuasive against the §101 rejection?

Answer: B

This is a classic exam trap: novelty (no prior art match) and "well-understood, routine, conventional" (§101/Step 2B) are different standards. Something can be technically novel yet still routine/conventional in the sense Step 2B cares about — a §102 argument does not automatically win a §101 argument.

### **Rule 7: Streamlined Eligibility Analysis (§2106.06)**

Streamlined eligibility analysis means:

Answer: B

It's procedural, not substantive — same Alice/Mayo test, just not fully spelled out in the office action when the answer is obvious. Don't mistake it for a separate/easier eligibility test.

## **Chapter Close-Out Quiz**

### **1. (B)**

A correlation between a biomarker level and a disease state is a law of nature / natural phenomenon — a naturally occurring relationship, not something invented.

### **2. (B)**

Merely observing/detecting a natural correlation, without any claimed step that acts on it (e.g., administering a treatment), doesn't integrate it into a practical application — classic Mayo v. Prometheus pattern.

### **3. (B)**

A specific, novel alloy with a defined crystalline structure not found in nature plainly isn't directed to an abstract idea or natural phenomenon on its face — a textbook streamlined-analysis candidate.

### **4. (A) True**

Eligibility (§101) and novelty (§102) are independent inquiries. Passing one doesn't guarantee passing the other, and a claim can clear §101 and still be rejected as anticipated under §102.

#### 5. (B)

"Hedging risk" is a commercial/legal interaction (organizing human activity); "calculating a fixed bill amount" is a mathematical concept. Claims often combine two groupings at once — don't assume only one applies.

#### 6. (B)

Transmitting/storing data is repeatedly cited in the guidance as a textbook example of insignificant extra-solution activity / well-understood, routine, conventional activity — it does not supply an inventive concept.

#### 7. (B)

A Step 1 category failure is a complete, standalone basis for rejection — you don't need to (and shouldn't) proceed to Step 2A or 2B once a claim fails Step 1.

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End of pilot section (MPEP §§2103–2106). Tell me what to adjust before I build out the rest of Chapter 2100 (§§2107 Utility, 2110s Prior Art/§102, 2140s Obviousness/§103, 2160s §112) in this same format.